

A Tool Developed in the Erasmus+ [Metacog](#) Project

MetaCheck Documentation

Complete guide to using MetaCheck for educational fact-checking and media literacy learning

GitHub Link: <https://github.com/umairalipathan1980/MetaCheck---An-Evidence-Based-Fact-Checker/tree/main>

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1. How to Use

Overview

MetaCheck is an educational tool that helps students learn fact-checking by comparing their own assessment with AI-generated analysis. The workflow consists of three main steps:

1. **Your Assessment** - Do your own fact-checking first
2. **AI Analysis** - Let AI verify the same claims
3. **Compare** - See how your assessment compares with AI

Step 1: Your Assessment

Why start with your assessment?

By doing your own fact-checking first, you develop critical thinking skills before seeing the AI's approach. This pedagogical design prevents "answer-seeking" behavior and encourages genuine learning.

How to create your assessment:

1. **Add Claims Manually**
 - Click "+ Add New Claim" button
 - Enter the claim text you want to verify
 - Research the claim using multiple sources
 - Gather evidence and assess credibility
2. **Determine Your Verdict**
 - Choose from: SUPPORTED, REFUTED, INSUFFICIENT_INFORMATION, or CONFLICTING_EVIDENCE
 - Set your confidence level (0.0 to 1.0)
 - Record how many sources you consulted
 - Track time spent on verification
3. **Write Your Reasoning**
 - Explain why you reached your verdict
 - Cite specific sources and evidence
 - Note any assumptions or uncertainties

Guidelines for Assessment:

Refer to the collapsible "How to Assess Claims (Guidelines)" section in the Your Assessment tab, which includes:

- **Extract Verifiable Claims:** Identify statements that can be proven true or false
- **Collect Evidence from Multiple Sources:** Use search engines, Wikipedia, fact-checking sites, primary sources
- **Evaluate Source Credibility:** Consider authority, bias, evidence quality, timeliness
- **Choose a Verdict:** Based on the evidence strength
- **Set Confidence Level:** How certain are you about your verdict?
- **Write Your Reasoning:** Explain your critical thinking process

Your Claims (1)
Claims you've created and assessed

Claim 1 **SUPPORTED** Confidence: 0.90

President Trump has made statements about taking large doses of aspirin to prevent heart disease.

I found exact snippets from reliable sources confirming this claim.

Sources: 3 | Time: 5m

Edit Claim
Update your claim and assessment

Claim Text (required)

President Trump has made statements about taking large doses of aspirin to prevent heart disease.

Verdict: **SUPPORTED** Confidence: 0.90

Sources checked: 3 Time spent (minutes): 5

Reasoning (required)

I found exact snippets from reliable sources confirming this claim.

Key sources (one per line)

Google Fact Check,
Snop,
Wikipedia

Update Claim **Cancel**

Add your own assessment for a verification task

Step 2: AI Analysis

Automated Verification:

After completing your assessment, you can run AI verification on the same text:

1. **Input Text:** Paste or type the text containing claims
2. **Extract Claims:** AI automatically identifies verifiable claims
3. **Select Claims:** Choose which claims to verify (up to 5)
4. **Run Verification:** AI searches multiple sources and generates verdicts

What AI Does:

- Searches 3 evidence sources in parallel (Web, Wikipedia, Fact Check databases)
- Assigns credibility scores to each source
- Analyzes stance (supports/refutes/neutral)
- Synthesizes a verdict with confidence and justification

- Provides full transparency through metacognitive detail (Comprehensive mode)

Basic vs. Comprehensive Mode:

- **Basic:** Faster, concise justifications (1-2 sentences)
- **Comprehensive:** Detailed reasoning, full metacognitive transparency, more sources

Step 3: Compare & Learn**Educational Comparison:**

After both assessments are complete:

1. View side-by-side comparison of your verdict vs. AI verdict
2. See overall summary of your performance
3. Review areas for improvement (if any)
4. Learn professional fact-checking techniques

What You'll Learn:

- How professional fact-checkers approach claims
- Source selection strategies
- Evidence synthesis techniques
- Confidence calibration
- Reasoning transparency

STEP 2 AI Analysis & Results

Run MetaCheck

Paste text with one or more factual claims. MetaCheck will extract, search, classify domains, and return verdicts with metacognitive details.

President Trump's statements about taking large doses of aspirin to prevent heart disease reflect medical guidelines

Max 2,000 characters

116 / 2,000

Mode: Basic Comprehensive

Clear

Extract Claims

CLAIMS

1

AVG CONFIDENCE

0.86

ELAPSED

11489 ms

Generated at: 2026-02-08

Export JSON

Select Claims to Verify

2 claims extracted • 2 selected • Max 5 in basic mode

Select All

Deselect All

☒ President Trump has made statements about taking large doses of aspirin to prevent heart disease.

Claim #1

Worthiness: 78%

☒ Taking large doses of aspirin to prevent heart disease reflects current medical guidelines.

Claim #2

Worthiness: 92%

Ready to verify 2 claims

Verify Selected Claims

AI assessment (view 1)

Claims overview

Basic mode shows essential facts. Switch to comprehensive for full reasoning.

SUPPORTED

Claim 1

President
Trump has
made...

Confidence
0.86

Sources: 5

SUPPORTED

Confidence 0.86

MODE: BASIC

President Trump has made statements about taking large doses of aspirin to prevent heart disease.

Multiple credible reports state that Donald Trump has repeatedly said he takes a large or adult-dose aspirin daily, more than his doctors recommend, to thin his blood and prevent cardiovascular disease.

Basic mode shows essential facts only. Re-run in comprehensive mode to see full metacognitive reasoning.

Key Sources

- FactCheck.org – "Trump's Aspirin Use and Doctors' Recommendations"
- Cape Cod Healthcare – "President Trump says he takes a daily adult aspirin"
- YouTube news clip – "Trump says he's taking more aspirin than his doctors ..."
- Wikipedia: Paracetamol
- Wikipedia: Fentanyl

WEB_SEARCH | SUPPORTS

Cred: 0.60

Relevance: 1.00

Trump's Aspirin Use and Doctors' Recommendations

<https://www.factcheck.org/2026/01/trumps-aspirin-use-and-doctors-recommendations/>

SNIPPET

President Donald Trump has said on multiple occasions in recent months that he takes a "large" dose of aspirin to prevent cardiovascular disease

WEB_SEARCH | SUPPORTS

Cred: 0.60

Relevance: 1.00

President Trump says he takes a daily adult aspirin. Should ...

<https://www.capecodhealth.org/cape-cod-health-news/president-trump-says-he-takes-a-daily-adult-aspirin-should-you/>

SNIPPET

President Donald Trump recently said he takes a daily adult aspirin (325 mg) to thin his blood, despite medical advice that he should take a

WEB_SEARCH | SUPPORTS

Cred: 0.50

Relevance: 1.00

Trump says he's taking more aspirin than his doctors ...

<https://www.youtube.com/watch?v=eJzL5u4Y110>

SNIPPET

The president told the Wall Street Journal his excessive hand bruising is caused by the amount of aspirin he takes, saying it's more than his doctors recommend

AI assessment (view 2)

STEP 4 **Compare**

Comparison Summary

Overview of your claims vs AI-extracted claims

1

Your Claims

1

AI-Extracted Claims

100%

Verdict Agreement

Avg confidence diff: 0.04

Your Claims (1)

CLAIM 1
SUPPORTED

President Trump has made statements about taking large doses of aspirin to prevent heart disease.

Confidence: **0.90** Sources: 3

Time: 5m

Your Reasoning:
I found exact snippets from reliable sources confirming this claim.

Your Sources:

- Google Fact Check
- Snop
- Wikipedia

AI-Extracted Claims (1)

CLAIM 1
SUPPORTED
✓ MATCH

President Trump has made statements about taking large doses of aspirin to prevent heart disease.

Confidence: **0.86** Sources: 5

Comparison with Your Claim 1:
Confidence diff: 0.04
Verdict: **Agreement** ✓

AI Justification:
Multiple credible reports state that Donald Trump has repeatedly said he takes a large or adult-dose aspirin daily, more than his doctors recommend, to thin his blood and prevent cardiovascular disease.

AI Sources:

- FactCheck.org – "Trump's Aspirin Use and Doctors' Recommendations"
- Cape Cod Healthcare – "President Trump says he takes a daily adult aspirin"
- YouTube news clip – "Trump says he's taking more aspirin than his doctors ..."
- Wikipedia: Paracetamol
- Wikipedia: Fentanyl

AI Feedback & Analysis

Get personalized feedback on your fact-checking performance

Summary

Your verdict and understanding match the AI's exactly for this claim: you correctly identified it as supported and used multiple sources, which aligns well with the AI's analysis.

Comparison of user vs. AI assessment with feedback

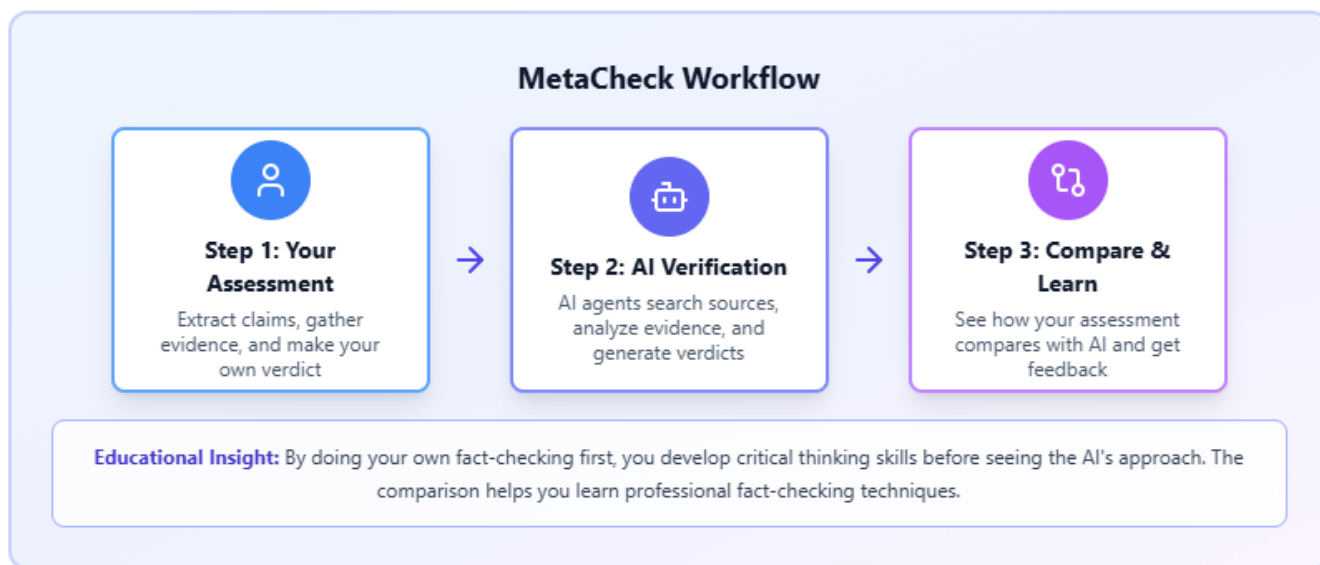
2. How It Works

MetaCheck Workflow

Educational Workflow:

1. **Your Assessment** → Extract claims, gather evidence, make your own verdict
2. **AI Verification** → AI agents search sources, analyze evidence, generate verdicts
3. **Compare & Learn** → See how your assessment compares with AI and get feedback

Educational Insight: By doing your own fact-checking first, you develop critical thinking skills before seeing the AI's approach. The comparison helps you learn professional fact-checking techniques.



How MetaCheck Works Behind the Scenes

Understanding the technical workflow helps you interpret results and troubleshoot issues. MetaCheck uses an **agentic AI architecture** with specialized components working together.

Complete Technical Workflow

Step 1: INPUT - User provides text to verify

Step 2: CLAIM EXTRACTION - AI extracts verifiable claims (Claim Extractor Agent)

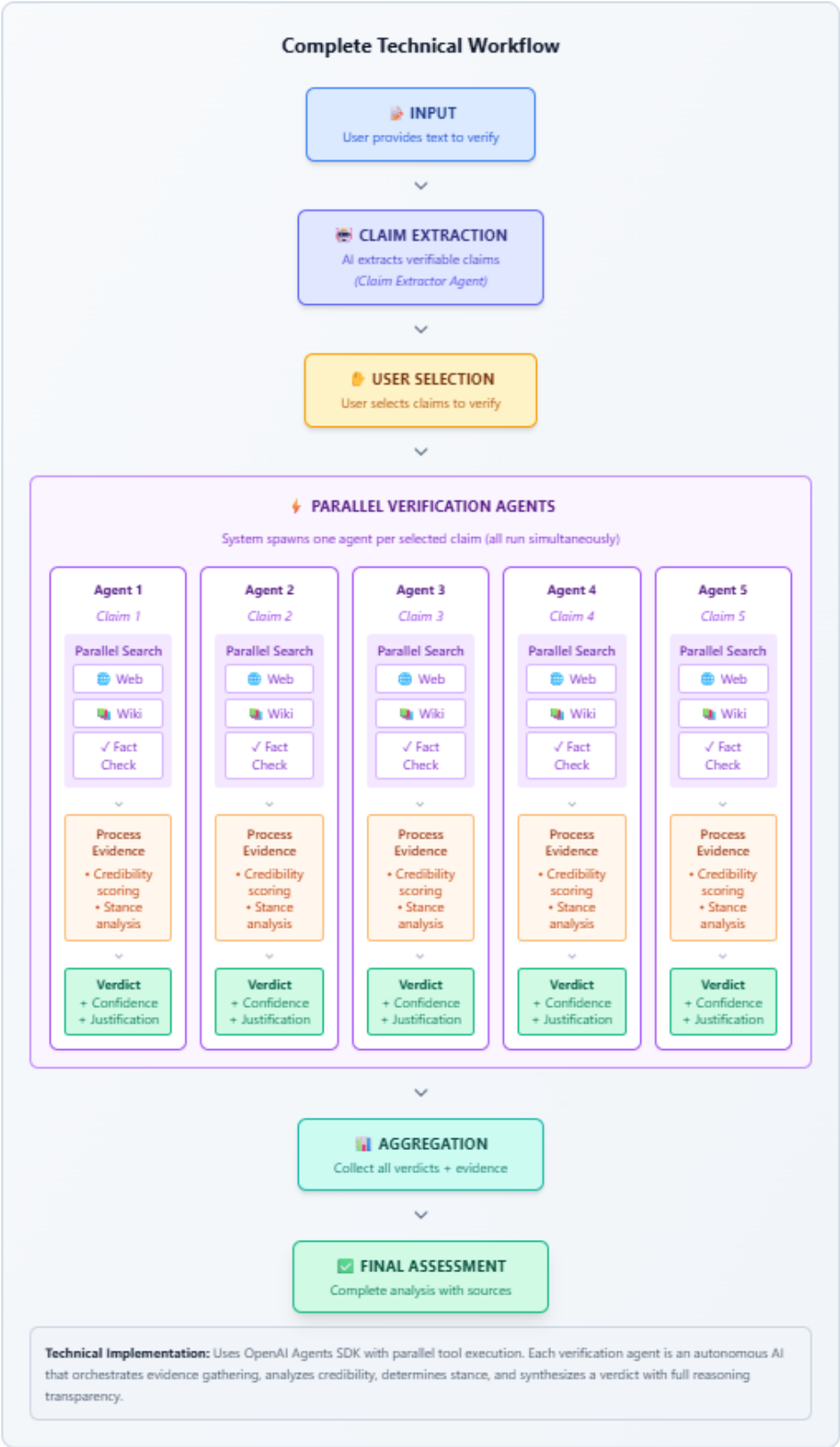
Step 3: USER SELECTION - User selects claims to verify

Step 4: PARALLEL VERIFICATION AGENTS - System spawns one agent per selected claim (all run simultaneously) - Each agent performs: - **Parallel Search:** Web 🌐, Wikipedia 📖, Fact Check ✓ - **Process Evidence:** Credibility scoring + Stance analysis - **Verdict:** With Confidence + Justification

Step 5: AGGREGATION - Collect all verdicts + evidence

Step 6: FINAL ASSESSMENT - Complete analysis with sources

Technical Implementation: Uses OpenAI Agents SDK with parallel tool execution. Each verification agent is an autonomous AI that orchestrates evidence gathering, analyzes credibility, determines stance, and synthesizes a verdict with full reasoning transparency.



How MetaCheck works behind the scenes

The Verification Pipeline

Phase 1: Claim Extraction

Component: Claim Extractor Agent

What happens: The AI analyzes your input text and identifies statements that are: - Specific and concrete (not vague) - Falsifiable (can be proven true or false) - Require external verification (not common knowledge) - Contain verifiable details (numbers, dates, names, locations)

Output: A list of candidate claims extracted from your text

Phase 2: Claim Selection

Component: User Interface (Manual Selection)

What happens: You review the extracted claims and select which ones you want to verify. This gives you control over what gets fact-checked and helps manage processing time and cost.

Phase 3: Parallel Evidence Gathering

Component: Orchestrator Agent with comprehensive_evidence_tool

What happens: For each selected claim, the system simultaneously searches three sources:

1. **Tavily Web Search:** Finds relevant web pages and snippets
2. **Wikipedia:** Searches encyclopedic articles
3. **Google Fact Check API:** Queries professional fact-checking databases

All three searches run **in parallel** (simultaneously), significantly reducing processing time.

Output: Combined evidence bundle with snippets, URLs, source types, and preliminary scores

Phase 4: Credibility Classification

Component: Domain Classification (config-based lookup)

What happens: Each web search result is analyzed to determine its credibility: - Domain is extracted from the URL - Domain is matched against a taxonomy (government, academic, news, etc.) - Credibility score is assigned based on category (0.50-0.90)

Note: Wikipedia (0.80) and Fact Check (0.95) have fixed scores. Only web search results use dynamic domain classification.

Phase 5: Stance Analysis

Component: Orchestrator Agent (AI Analysis)

What happens: The AI reads each evidence snippet and determines its stance: - **Supports:** Evidence confirms or aligns with the claim - **Refutes:** Evidence contradicts or disproves the claim - **Neutral:** Evidence is related but doesn't clearly support or refute - **Unclear:** Evidence is ambiguous or indirect

For Fact Check sources, the stance is automatically interpreted from ratings. For web and Wikipedia, the AI analyzes the snippet text.

Phase 6: Verdict Synthesis

Component: Orchestrator Agent (Decision Logic)

What happens: The AI synthesizes all evidence and reaches a verdict:

- 1. Calculate total credibility weight for supporting evidence
- 2. Calculate total credibility weight for refuting evidence
- 3. Apply verdict criteria based on weights and thresholds
- 4. Determine confidence score based on evidence quality and quantity
- 5. Generate justification explaining the verdict
- 6. (Comprehensive mode only) Document metacognitive detail with full reasoning transparency

Output: VerificationResult with verdict, confidence, justification, sources, and evidence list

Mode Differences Explained

Aspect	Basic Mode	Comprehensive Mode
Primary Use Case	Quick fact-checking, classroom demos, fast results	Deep learning, understanding reasoning, and educational transparency
Evidence Retrieved	Limited (3 web, 2 wiki, 2 fact-check)	More comprehensive (5 web, 3 wiki, 3 fact-check)
Justification	1-2 concise sentences explaining the verdict	Detailed paragraph with reasoning and evidence summary
Metacognitive Detail	Not included (empty)	Full transparency: search queries, source assessments, verdict reasoning, AI uncertainties, assumptions
Processing Time	Faster (~10-20 seconds per claim)	Slower (~15-30 seconds per claim)
Max Claims	5 claims	5 claims
Best For	Quick demos, time-constrained situations	Educational deep dives, learning AI reasoning, and research
API Cost	Lower (fewer API calls)	Higher (more comprehensive analysis)

Parallelism Architecture

MetaCheck uses **two levels of parallelism** for optimal performance:

- 1. **Agent-Level Parallelism:** All selected claims are verified simultaneously (not sequentially)
- 2. **Tool-Level Parallelism:** Within each agent, all 3 evidence sources are searched in parallel

This architecture significantly reduces total verification time.

3. Understanding Verdicts

Verdict Categories

MetaCheck uses four verdict categories based on evidence strength and consistency:

SUPPORTED

Meaning: The claim is backed by credible evidence from multiple sources.

Criteria: - Supporting evidence significantly outweighs refuting evidence - High credibility sources confirm the claim - Consensus across multiple independent sources - No significant contradictions from reliable sources

Example: “The Eiffel Tower is 330 meters tall” → Multiple authoritative sources (Wikipedia, official websites, encyclopedias) confirm this measurement.

Confidence Interpretation: - 0.8-1.0: Very strong support from many high-quality sources - 0.6-0.7: Moderate support, some gaps or lower-quality sources - 0.5: Borderline, minimal support

REFUTED

Meaning: The claim is contradicted by credible evidence.

Criteria: - Refuting evidence significantly outweighs supporting evidence - High credibility sources contradict the claim - Fact-checkers have rated it false or misleading - Evidence demonstrates the opposite is true

Example: “Vaccines cause autism” → Multiple scientific studies, medical organizations, and fact-checkers have thoroughly debunked this claim.

Confidence Interpretation: - 0.8-1.0: Very strong refutation from authoritative sources - 0.6-0.7: Clear contradiction but from fewer sources - 0.5: Borderline, weak refutation

CONFLICTING EVIDENCE

Meaning: Different credible sources provide contradictory information.

Criteria: - Both supporting and refuting evidence exist in significant amounts - Credible sources disagree with each other - No clear consensus in the evidence - The truth may depend on context, interpretation, or ongoing debate

Example: “Remote work increases productivity” → Some studies show productivity gains, others show productivity losses. The truth varies by industry, role, and measurement methodology.

When to use: - Genuine scientific or expert disagreement - Politically polarized topics with partisan sources - Evolving situations where facts are still being established - Complex claims where “it depends” is the honest answer

? INSUFFICIENT_INFORMATION

Meaning: Not enough reliable evidence was found to make a determination.

Criteria: - Very few sources found (or sources have low credibility scores) - Total credibility weight below minimum threshold (typically < 0.7) - Evidence is tangential or doesn't directly address the claim - Sources are unreliable, outdated, or irrelevant

Example: "John Smith's bakery in Springfield has the best croissants" → Too specific/local to have authoritative sources. Subjective claim without verifiable evidence.

Common Causes: - Very new events (not yet in databases) - Hyper-local claims (small town events, local businesses) - Obscure historical facts - Claims about non-public figures - Subjective statements ("best", "most beautiful")

Confidence Scores

What is confidence?

A numerical score (0.0 to 1.0) indicating how certain the AI is about its verdict.

How it's calculated:

Confidence is based on: - **Evidence quantity:** More sources = higher confidence - **Evidence quality:** Higher credibility sources = higher confidence - **Evidence consistency:** Agreement among sources = higher confidence - **Search success:** Successful searches = higher confidence

Interpretation:

- **0.9-1.0:** Extremely confident (strong, consistent evidence from many high-quality sources)
- **0.7-0.8:** Confident (solid evidence from reliable sources)
- **0.5-0.6:** Moderate confidence (some evidence, but gaps or inconsistencies)
- **0.3-0.4:** Low confidence (weak evidence, few sources, low quality)
- **0.0-0.2:** Very low confidence (almost no evidence, unreliable sources)

Important Notes:

- High confidence does NOT guarantee the verdict is correct (AI can be confidently wrong)
- Low confidence often indicates insufficient evidence, not necessarily that the claim is false
- Confidence reflects evidence strength, not absolute truth

Justification

What is justification?

A human-readable explanation of why the AI reached its verdict. This is the "reasoning" section that explains the evidence and decision process.

Basic Mode Justification: - 1-2 concise sentences - Summarizes the key evidence - States the verdict and primary reason

Comprehensive Mode Justification: - Detailed paragraph - Explains evidence from each source
- Discusses evidence weight and credibility - Addresses contradictions or uncertainties -
Transparent about reasoning process

Metacognitive Detail (Comprehensive Mode Only)

What is metacognitive detail?

A detailed breakdown of the AI's reasoning process, showing "thinking out loud" transparency.

What's included:

1. **Search Queries Used:** Exact queries sent to each evidence source
2. **Source Assessments:** How each source was evaluated for credibility and stance
3. **Evidence Synthesis:** How different pieces of evidence were combined
4. **Verdict Reasoning:** The logical steps that led to the final verdict
5. **Uncertainties & Assumptions:** What the AI is unsure about or had to assume
6. **Alternative Interpretations:** Other ways the evidence could be interpreted

Why is it useful?

- **Educational:** Students learn how professional fact-checkers think
- **Transparency:** Users can audit the AI's reasoning
- **Debugging:** Helps identify when AI misunderstood something
- **Trust:** Users can verify the reasoning makes sense

Evidence Sources

Each verdict includes a list of evidence sources with:

- **URL:** Link to the source
- **Title:** Title or snippet from the source
- **Source Type:** web, wikipedia, or factcheck
- **Credibility Score:** 0.0-1.0 (higher = more trustworthy)
- **Stance:** supports, refutes, neutral, or unclear
- **Domain Category:** government, academic, news, etc. (for web sources)

Search Status

Each evidence source also has a search status:

- **success:** Search ran successfully and found results
- **no_results:** Search completed but found no matching evidence
- **error:** Search failed due to API error or network issue
- **no_api_key:** API key not configured for this source type

Common Verdict Patterns

Pattern 1: Strong Consensus - Verdict: SUPPORTED or REFUTED - Confidence: 0.8-1.0 - All sources agree - High credibility sources

Pattern 2: Weak Evidence - Verdict: INSUFFICIENT_INFORMATION - Confidence: 0.3-0.5 - Few sources found - Low credibility scores

Pattern 3: Genuine Disagreement - Verdict: CONFLICTING_EVIDENCE - Confidence: 0.5-0.7 - Credible sources on both sides - No clear consensus

Pattern 4: Search Failures - Verdict: INSUFFICIENT_INFORMATION - Confidence: 0.0-0.3 - Multiple search errors - No usable evidence retrieved

4. Evidence Sources

Overview of Evidence Sources

MetaCheck searches three types of evidence sources for each claim:

1. **Web Search (Tavily)**
2. **Wikipedia**
3. **Google Fact Check API**

All three sources are queried **in parallel** (simultaneously) to maximize speed.

1. Web Search (Tavily)

What is Tavily?

Tavily is a search API optimized for AI agents and LLM applications. It provides relevant, high-quality web search results with clean snippets.

How MetaCheck uses it:

- Searches the open web for claim-related content
- Returns relevant snippets and URLs
- Prioritizes authoritative sources
- Filters out low-quality content

Search Depth:

- **Basic:** Quick search, fewer results
- **Advanced:** More thorough search, more results (configurable in Settings)

Credibility Scoring:

Web search results are scored based on **domain classification**:

Domain Category	Credibility Score	Examples
Government	0.90	.gov, .mil, official

Domain Category	Credibility Score	Examples
Academic	0.85	government sites .edu, university sites, research institutions
Medical/Health Authority	0.85	WHO, CDC, NIH, medical journals
Fact-Checking Organizations	0.85	Snopes, FactCheck.org, PolitiFact
Major News (High Reputation)	0.75	AP, Reuters, BBC, NYT, WSJ
Major News (Mainstream)	0.70	CNN, Fox News, Guardian, etc.
Non-Profit Organizations	0.70	.org domains, NGOs
General News/Media	0.60	Regional news, smaller outlets
Commercial/Business	0.55	.com domains, corporate sites
Unknown/Uncategorized	0.50	Domains not in taxonomy

Limitations:

- May miss paywalled content
- Cannot access private databases
- Subject to search API coverage
- Domain classification may not capture all nuances

2. Wikipedia

What is Wikipedia?

Wikipedia is a free online encyclopedia with millions of articles. While not peer-reviewed, it's generally reliable for factual information and provides good starting points for research.

How MetaCheck uses it:

- Searches Wikipedia articles for claim-related content
- Extracts relevant paragraphs
- Provides context and background information
- Links to original articles for verification

Credibility Score: Fixed at **0.80**

Why 0.80? - Wikipedia is generally accurate for mainstream topics - Articles are collaboratively edited and monitored - Citations provide traceability to original sources - Not peer-reviewed, so not as authoritative as academic sources

Strengths: - Comprehensive coverage of many topics - Well-structured, neutral point of view - Frequently updated - Good for historical facts, definitions, basic information

Limitations: - Can be vandalized (though usually quickly corrected) - Not appropriate for highly specialized topics - May lag behind very recent events - Not suitable as a primary source for academic research

3. Google Fact Check API

What is Google Fact Check API?

The Google Fact Check Tools API searches a database of fact-checks from professional fact-checking organizations worldwide.

How MetaCheck uses it:

- Queries the fact-check database for matching claims
- Returns fact-checks from organizations like Snopes, FactCheck.org, PolitiFact
- Includes the fact-checker's verdict (e.g., "True", "False", "Mostly False")
- Provides claim context and reasoning

Credibility Score: Fixed at 0.95

Why 0.95? - Professional fact-checkers are experts in verification - Rigorous editorial standards - Transparent methodology - Explicitly designed for claim verification

Strengths: - Highest credibility of all sources - Expert analysis with detailed reasoning - Covers popular claims and misinformation - Includes context and explanation

Limitations: - Coverage is limited to claims that fact-checkers have investigated - Typically focuses on viral misinformation and public statements - May not cover obscure or local claims - No results ≠ claim is false (just means not fact-checked yet)

Source Comparison

Aspect	Web Search (Tavily)	Wikipedia	Fact Check API
Credibility	0.50-0.90 (varies)	0.80 (fixed)	0.95 (fixed)
Coverage	Broad (entire web)	Encyclopedic topics	Fact-checked claims only
Freshness	Very current	Usually current	Depends on fact-checkers
Depth	Varies widely	Medium depth	Deep analysis
Expertise	Varies by source	Collaborative	Professional fact-checkers
Best For	Current events, diverse perspectives	Background, definitions, history	Debunked claims, misinformation

Evidence Synthesis

How MetaCheck combines evidence:

1. **Collect** all evidence from all three sources
2. **Score credibility** based on source type and domain
3. **Determine stance** (supports/refutes/neutral) for each piece of evidence
4. **Calculate weighted evidence** (credibility × stance)
5. **Synthesize verdict** based on total evidence weight
6. **Generate confidence** based on quantity, quality, and consistency

Search Status Indicators

Each source search has a status:

Status	Meaning	Implication	Action
success	Search ran successfully and found results	Evidence is usable	None — working as expected
no_results	Search ran but found no matching evidence	This is an absence of evidence, not a system error	None — this is normal for many claims
error	Search failed due to API error, network issue	Evidence may be incomplete. Verdict is based only on sources that succeeded	Try again later or check system status
no_api_key	API key for this source was not configured	That source type was disabled for this run. Verdict is based on remaining sources	System administrator needs to configure API key

Quality Assurance

How to assess evidence quality:

1. **Check credibility scores:** Higher is better (> 0.7 is generally reliable)
2. **Verify source diversity:** Multiple independent sources are better than one
3. **Review search status:** Ensure searches succeeded
4. **Read justification:** Understand how AI interpreted the evidence
5. **Follow links:** Verify AI interpretation by reading original sources
6. **Check for bias:** Consider potential biases in source selection

5. Educational Use

Learning Objectives

MetaCheck is designed to help students develop:

1. **Critical Thinking:** Evaluate claims independently before seeking answers
2. **Media Literacy:** Understand how to assess source credibility
3. **Research Skills:** Learn effective search strategies and source evaluation
4. **Metacognition:** Reflect on their own reasoning process
5. **AI Literacy:** Understand how AI makes decisions and where it can fail

Pedagogical Approach

Compare-and-Contrast Learning:

MetaCheck uses a “Do-it-yourself first, then compare” pedagogy:

1. **Student completes task independently** (no AI guidance)
2. **AI completes the same task** (transparent reasoning)
3. **Student compares** their approach with AI’s approach
4. **Student reflects** on differences and learns from comparison

Why this works:

- **Prevents answer-seeking:** Students must think first
- **Encourages genuine effort:** No way to “cheat” by copying AI
- **Promotes metacognition:** Students see their own thinking vs. expert thinking
- **Builds confidence:** Students see what they got right
- **Identifies gaps:** Students see where they can improve

Classroom Use Cases

Use Case 1: Fact-Checking Exercise

Objective: Teach students how to verify claims using multiple sources

Activity: 1. Provide a text with several factual claims (news article, social media post) 2. Students manually fact-check using search engines and databases 3. Students record their findings in MetaCheck 4. Run AI verification on the same text 5. Compare results and discuss differences in class

Discussion Points: - Which sources did students miss? - How did students assess credibility differently than AI? - What search strategies were most effective?

Use Case 2: Misinformation Detection

Objective: Help students identify and debunk common misinformation

Activity: 1. Present a viral claim or conspiracy theory 2. Students research and create their own assessment 3. AI verifies the same claim 4. Compare verdicts and review fact-checker sources 5. Discuss why misinformation spreads despite evidence

Discussion Points: - What makes a source credible? - Why do people believe false claims? - How can we communicate corrections effectively?

Use Case 3: Source Credibility Training

Objective: Teach students to evaluate source reliability

Activity: 1. Students assess a claim with diverse sources (academic, news, blogs, social media) 2. Students assign their own credibility scores to each source 3. Compare with MetaCheck's domain classification system 4. Discuss agreements and disagreements

Discussion Points: - What makes a source credible? - How should domain (.gov, .edu, .com) influence trust? - When might a "lower credibility" source still be valuable?

Use Case 4: AI Transparency & Limitations

Objective: Help students understand how AI works and where it can fail

Activity: 1. Give students a tricky claim (ambiguous, context-dependent, or nuanced) 2. Students predict how AI will handle it 3. Run verification in Comprehensive mode 4. Analyze metacognitive detail 5. Identify where AI succeeded and where it struggled

Discussion Points: - What can AI do well vs. poorly? - How can we audit AI reasoning? - When should we trust AI vs. human judgment?

Assessment Rubrics

Suggested Grading Criteria for Student Assessments:

Criterion	Poor (0-2)	Satisfactory (3-4)	Excellent (5)
Evidence Quality	Used 0-1 sources, unreliable sources	Used 2-3 sources, mostly credible	Used 3+ diverse, highly credible sources
Source Evaluation	Did not assess credibility	Basic credibility assessment	Thorough credibility analysis with reasoning
Reasoning	Vague or missing justification	Clear reasoning with some evidence	Detailed, logical reasoning citing specific evidence
Verdict Accuracy	Incorrect verdict based on evidence	Reasonable verdict but could be stronger	Well-supported verdict matching evidence
Confidence Calibration	Confidence wildly	Confidence somewhat aligned	Confidence well-calibrated to evidence

Criterion	Poor (0-2)	Satisfactory (3-4)	Excellent (5)
	misaligned with evidence		strength

Tips for Instructors

Before using MetaCheck:

1. **Set expectations:** Explain the “do-it-yourself first” approach
2. **Model the process:** Demonstrate a complete assessment as an example
3. **Provide guidelines:** Share the assessment guidelines from the tool
4. **Discuss credibility:** Review the credibility scoring system
5. **Practice:** Start with simple, unambiguous claims

During the activity:

1. **Encourage thoroughness:** Remind students to use multiple sources
2. **Require documentation:** Students should record sources and reasoning
3. **Allow adequate time:** Fact-checking takes time (15-30 min per claim)
4. **Discourage guessing:** Better to say “insufficient information” than guess
5. **Monitor progress:** Check that students are engaging meaningfully

After comparison:

1. **Facilitate reflection:** Have students write about what they learned
2. **Discuss patterns:** Identify common mistakes or successes
3. **Address surprises:** Talk about unexpected differences
4. **Highlight strengths:** Celebrate what students did well
5. **Identify improvements:** Discuss specific areas for growth

Advanced Features for Educators

Settings Configuration (Admin Only):

Instructors can adjust: - Number of claims to verify per run - Number of sources to search - Model choice (faster vs. more accurate) - Search depth (basic vs. advanced) - Timeout settings

Comparison Feedback:

The comparison tab provides: - Overall summary of student performance - Areas for improvement (specific, actionable feedback) - Recognition of what was done well

Export Functionality:

Students and instructors can export: - Complete assessment as JSON - Full verification results - Comparison analysis

Best Practices

For Students:

1. **Take your time:** Thorough fact-checking requires patience
2. **Use multiple sources:** Don't rely on just one website
3. **Check credibility:** Look for author credentials, publication date, domain
4. **Be honest about confidence:** It's okay to be uncertain
5. **Cite specific evidence:** Don't just say "I googled it"
6. **Document your process:** Track sources, time, reasoning

For Instructors:

1. **Start simple:** Use clear, unambiguous claims first
2. **Scaffold the process:** Provide templates or worksheets initially
3. **Make it collaborative:** Have students work in pairs or small groups
4. **Connect to current events:** Use recent news or viral claims
5. **Emphasize learning over grades:** Focus on improvement, not perfection
6. **Model transparency:** Share your own fact-checking process

Common Challenges & Solutions

Challenge 1: Students rush through assessment

Solution: - Require minimum number of sources (3-5) - Have students submit source list before verdict - Track and require minimum time spent - Make assessment a significant portion of grade

Challenge 2: Students copy AI results

Solution: - Require students to submit assessment BEFORE running AI - Use timestamp tracking - Require written reflection on differences - Focus on process, not just final answer

Challenge 3: Disagreement with AI verdict

Solution: - Emphasize that disagreement is okay (AI can be wrong) - Have students defend their position with evidence - Discuss nuances and context the AI may have missed - Use it as a teaching moment about AI limitations

Challenge 4: Students don't know where to start

Solution: - Provide assessment guidelines (included in tool) - Model the process with an example - Offer search strategy tips - Start with simpler, more straightforward claims

Resources for Further Learning

Fact-Checking Organizations: - [FactCheck.org](#) - [Snopes](#) - [PolitiFact](#) - [Full Fact \(UK\)](#)

Media Literacy Resources: - [News Literacy Project](#) - [Common Sense Media](#) - [NAMLE \(National Association for Media Literacy Education\)](#)

Source Evaluation Tools: - [CRAAP Test](#) (Currency, Relevance, Authority, Accuracy, Purpose) - [SIFT Method](#) (Stop, Investigate source, Find better coverage, Trace claims) - [Lateral Reading](#) (Stanford approach)

Appendix: Technical Specifications

System Requirements

Backend: - Python 3.12+ - FastAPI framework - OpenAI Agents SDK - Tavily API key - Google Fact Check API key (optional) - Wikipedia API access token (optional)

Frontend: - Node.js 20+ - React 19+ - Vite build tool - Tailwind CSS

API Endpoints

- POST /api/extract - Extract claims from text
- POST /api/verify - Verify selected claims
- POST /api/compare - Compare student vs AI assessment
- GET /api/config/settings - Get current settings
- PUT /api/config/settings - Update settings (admin only)
- POST /api/admin/reload-settings - Reload configuration (admin only)

Authentication

Admin access required for: - Modifying settings - Reloading configuration

Login credentials: - Username: admin - Password: Set in .env file (ADMIN_PASSWORD)

Configuration

Settings can be adjusted in the Settings tab (admin login required):

- **Max claims to verify per run:** 1-5
- **Max claims to extract:** 1-10
- **Max web search results:** 1-5
- **Max Wikipedia results:** 1-5
- **Max fact-check results:** 1-5
- **Model choice:** gpt-5.1, gpt-4.1-mini, or gpt-5-mini
- **Tavily search depth:** basic or advanced
- **Per-claim timeout:** seconds to wait before timing out

About MetaCheck

Version: 1.0.0

Developer: Educational AI Research Team

Purpose: MetaCheck is designed as an educational tool to help students learn fact-checking, critical thinking, and media literacy skills through hands-on practice and AI-powered comparison.

License: Educational use

Contact: For questions, issues, or feedback about MetaCheck, please contact your course instructor or system administrator.

This documentation reflects the MetaCheck interface as of the current version. Features and functionality may be updated over time. Always refer to the in-application documentation for the most current information.